

Vittanala Sai Kushal

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Education

Vellore Institute of Technology

Andhra Pradesh, India

B.Tech in Computer Science and Engineering (Specialization in Data Analytics) | CGPA: 8.56/10 Jul 2022 – Jun 2026

- **Coursework:** Data Structures & Algorithms, FDA, Operating Systems, OOP, Software Engineering, Computer Networks

Technical Skills

Programming Languages: Python, Java, SQL

ML/AI Frameworks: TensorFlow, Random Forest, KNN, OpenCV

Data Analysis & Visualization: Pandas, NumPy, Power BI, Tableau, Excel

Databases: MySQL, MongoDB

Version Control: Git, Github

Cloud & DevOps: AWS, Docker

Development Tools: VS Code, Jupyter Notebook, Google Colab

Internship

Brightix IT Solutions

Remote

Data Analyst Intern

Jul 2025 – Sep 2025

- Analyzed and cleaned 20,000+ retail sales records using SQL, Excel, and Python (Pandas), handling missing values, duplicates, and data inconsistencies.
- Developed interactive Power BI and Tableau dashboards for the Super Store Sales Analysis project, tracking KPIs such as revenue, profit, customer segments, and regional performance, reducing manual reporting by 60%.
- Derived actionable insights by identifying top-performing products, loss-making categories, and regional sales trends to support data-driven business decisions.

Projects

Hybrid Quantum-Classical Parkinson's Disease Detection using Explainable AI

Dec 2025 – Apr 2026

- Developed a hybrid quantum-classical Parkinson's disease prediction system using speech biomarkers and Quantum Machine Learning.
- Implemented QSVM, VQC, XGBoost, and SVM models using Qiskit, reducing 700+ features to 10 key biomarkers.
- Built a React + FastAPI platform with SHAP-based explainability for interpretable predictions.

Technologies: Python, TensorFlow, FastAPI, React, Pandas, QML, ML

Urban Traffic Congestion System

Aug 2025 – Nov 2025

- Built a traffic congestion prediction application using CNN-LSTM and CNN-GRU models, selecting the best-performing approach for accurate predictions.
- Processed traffic data with 30+ input features including time, location, and weather using Pandas.
- Implemented a web interface using Flask and Next.js to visualize congestion across 5+ locations.

Technologies: Python, TensorFlow, Flask, Next.js, Pandas

Research Publications

Prediction of Kidney Disease and Urinary Disease using Machine Learning

- Co-authored and presented a research paper at ADSSS Conference 2024 titled “Prediction of Kidney Disease and Urinary Disease using Machine Learning”, achieving 92.31% classification accuracy.
- Preprocessed and analyzed 3000+ patient medical records by imputing missing values, resolving feature inconsistencies, and standardizing clinical features for machine learning models.

Certifications

- **Oracle Certified:** OCI 2025 Generative AI Professional
- **AWS Certified:** Cloud Practitioner